MIL-A-52839(ME)
27 January 1975

MILITARY SPECIFICATION

ALARM SET, ANTI-INTRUSION, RESTRICTED AREA:

MINISID III, AN/GSQ-154

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers a seismic anti-intrusion detection device.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-636	- Boxes, Shipping, Fiberboard.

Military

MIL-P-116	- Preservation-Packaging, Methods of.
MIL-W-27076	- Workmanship Standards for Ground Electronic and Associated Equipment.
MIL-C-45662	- Calibration System Requirements.
MIL-D-52776	- Detector, Anti-Intrusion: DT-509/GSQ (MAGID T-4).
MIL-D-52778	- Detector, Anti-Intrusion: DT-383/GSQ (AAU).

STANDARDS

Military

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.

FSC 6350

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MIL-STD-454	- Standard General Requirements for Electronic Equipment.
MIL-STD-461	- Electromagnetic Interference Characteristics Requirements for Equipment.
MIL-STD-781	- Reliability Tests: Exponential Distribution.
MIL-STD-810	- Environmental Test Methods.

DRAWINGS

ME

TA13218E8090	- Alarm Set, Anti-Intrusion Restricted Area: MINISID III, AN/GSQ-154.

(Copies of specifications, standards, and drawings required by the suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

UNIFORM CLASSIFICATION COMMITTEE, AGENT

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, ATTN: Tariff Publishing Officer, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC., AGENT

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Associations, Inc., ATTN: Tariff Order Section, 1616 P Street, NW, Washington, DC 20036.)

3. REQUIREMENTS

3.1 Description. The Alarm Set, Anti-Intrusion Restricted Area: MINISID III, AN/GSQ-154 (hereinafter called MINISID III) shall be in accordance with Top Assembly TA13218E8090 and as specified herein. The MINISID III shall consist of a sensor case and unique electronics (hereinafter called SCUE), an

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encoder, transmitter, and battery. Unless otherwise specified (see 6.2), only the SCUE shall be furnished. When specified (see 6.2) the complete MINISID III shall be furnished. When the complete MINISID III is to be furnished, the Government will furnish the encoder, transmitter, and battery (see 3.11). When only the SCUE is furnished, the Government will lend encoders, transmitters, and batteries (see 3.12) in the quantities specified (see 6.2) to facilitate testing the SCUE.

3.1.1 Drawings. The drawings forming a part of this specification are end-product drawings. No deviation from the prescribed materials, dimensions or tolerances is permissible without prior approval of the contracting officer. Any data (e.g. shop drawings, layouts, flow sheets, processing procedures, etc.) prepared by the supplier or obtained from a vendor to support fabrication and manufacture of the production item shall be made available upon request for inspection by the contracting officer or his designated representative.

3.2 First article (preproduction model). The supplier shall furnish one or more MINISID III for examination and testing within the time frame specified (see 6.2) to prove prior to starting production that his production methods will produce MINISID III that comply with the requirements of this specification. Examination and tests shall be as specified in Section 4. Unless otherwise specified herein, all examinations and tests shall be conducted by the supplier subject to surveillance and approval by the Government (see 6.3). When specified (see 6.2), the Government will conduct any or all of the preproduction examinations and tests as specified (see 6.2).

3.3 Initial production. When specified (see 6.2), the supplier shall furnish to the Government one or more MINISID III for inspection as specified in 4.4.

3.4 Material. Material shall be as specified herein and as shown on the applicable drawings. Nonfungus nutrient materials shall be used where applicable.

3.5 Physical and environmental requirements. The MINISID III shall function as specified herein without damage (6.7.1) after being subjected to shock; vibration; temperature, humidity, and altitude extremes; salt fog; water immersion; and fungus as may be encountered in normal field usage when tested in accordance with the following:

(a) Vibration (4.6.3.11.1).
(b) Shock (4.6.3.11.2).
(c) High/Low temperature (4.6.3.12.1).
(d) High temperature (4.6.3.12.2).

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(e) Low temperature (4.6.3.12.3).
(f) Temperature shock (4.6.3.12.1).
(g) Altitude (4.6.3.12.4).
(h) Humidity (4.6.3.12.5).
(i) Salt fog (4.6.3.12.6).
(j) Immersion (4.6.3.12.7).
(k) Fungus growth (4.6.3.12.8).

3.5.1 Humidity. There shall be no corrosion which could in any manner affect operational, service or maintenance requirements.

3.5.2 Fungus growth. There shall be no evidence of viable fungus growth on any component or subassembly.

3.5.3 Salt fog. There shall be no corrosion or binding of moving parts.

3.5.4 Immersion. There shall be no evidence of leakage into the test item.

3.6 Activation sequence. An activation sequence shall be initiated as a result of the seismic disturbance of the geophone. This seismic signal (geophone output) shall be amplified and processed by the seismic module. The output of the seismic module shall be a positive pulse which causes the encoder to supply power to the transmitter. The encoder shall then clock out the split phase message format. A code pulse at the end of the message shall remove power from the transmitter and encoder.

3.6.1 Ancillary devices. The MINISID III shall operate in conjunction with ancillary devices. The MINISID III shall transmit an alarm as a result of an encoder call-up from the ancillary device only or from a combination of inputs from the MINISID III seismic module and the ancillary device. When the ancillary device is connected to the MINISID III, automatic switch-over shall be accomplished by a change from "the MINISID III only mode" to "the ancillary mode". The type of ancillary mode selected shall be a function of the circuitry in the ancillary device. Also, when the complementary identification (CID) control (see 3.9.3) on the SCUE is set in the "0" position, an encoder call-up from an ancillary device shall cause the MINISID III to transmit a message with the Identification (ID) set on the code plug. When set in the "1" position, the ID shall be the complement of the ID set on the code plug. The seismic module ID shall always be the one set on the code plug assembly regardless of the setting of the CID. When the MINISID III is operated in conjunction with an ancillary device, the interconnecting wiring shall be as shown in Figure 1.

3.7 Encoder. The encoder shall generate switching and timing functions necessary to control the structure and transmission of a message. This

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message shall be formed by the combination of the preamble, frame mark, message ID, sensor ID and parity, see Figures 2A and 2B. This combination of parts forms an "Identity Message" and is transmitted in response to an activation from the seismic module. The 18 bit message shall be transmitted digitally as split phase FSK at a bit rate of 300 or 75 bits per second (bps). The bit structure for the 300 bps and 75 bps bit rate shall be as shown in Figure 3. The following parenthetical values apply to the 75 bps bit rate. The deviation from center frequency shall be plus 3.0 kilohertz (kHz) ± 10 percent for a "mark" and minus 3.0 kHz ± 10 percent for a "space". The bit structure as portrayed in Figure 3 shall be as follows: A "one" shall be formed by 1.667 millisecond (ms) (6.67 ms) on the mark frequency, followed by 1.667 ms (6.67 ms) on the space frequency; a "zero" shall be formed by 1.667 ms (6.67 ms) on the space frequency followed by 1.667 ms (6.67 ms) on the mark frequency. Rise and fall time between mark and space transition shall be between 100 and 300 microseconds. The overshoot during transition shall not exceed the tolerance on the frequency deviation "mark" or "space". Long term bit rate stability shall be at least ± 3 percent and short term stability shall be ± 0.1 percent or better. The encoder shall contain a voltage regulator that shall be capable of delivering 55 milliamperes (ma) at 6.3 volts direct current (VDC) ± 0.2 VDC, of which a maximum of 25 milliwatts (mw) will be required by the encoder. The remaining power shall be available for use in the other modules. The 6.3 VDC source shall not be used as a current sink. The total power requirements for the encoder are: For 6.3 VDC, 1 mw during the quiescent mode, and 25 mw during signal processing, for 28 VDC: 1/2 mw during the quiescent mode, and 500 mw during signal processing.

Table I. Sensor ID Settings

ID NO	BITS 4 5	6 7	8 9
1	01	01	01
2	01	01	10
3	01	01	11
4	01	10	01
5	01	10	10
6	01	10	11
7	01	11	01
8	01	11	10
9	01	11	11

ID NO	BITS 4 5	6 7	8 9
11	10	01	01
12	10	01	10
13	10	01	11
14	10	10	01
15	10	10	10
16	10	10	11
17	10	11	01
18	10	11	10
19	10	11	11

ID NO	BITS 4 5	6 7	8 9
21	11	01	01
22	11	01	10
23	11	01	11
24	11	10	01
25	11	10	10
26	11	10	11
27	11	11	01
28	11	11	10
29	11	11	11

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Table I. Sensor ID Settings (Cont'd)

ID NO	BITS 45	67	89
31	00	01	01
32	00	01	10
33	00	01	11
34	00	10	01
35	00	10	10
36	00	10	11
37	00	11	01
38	00	11	10
39	00	11	11

ID NO	BITS 45	67	89
41	01	00	01
42	01	00	10
43	01	00	11
44	10	00	01
45	10	00	10
46	10	00	11
47	11	00	01
48	11	00	10
49	11	00	11

ID NO	BITS 45	67	87
51	01	01	00
52	01	10	00
53	01	11	00
54	10	01	00
55	10	10	00
56	10	11	00
57	11	01	00
58	11	10	00
59	11	11	00

ID NO	BITS 45	67	89
61	00	00	01
62	00	00	10
63	00	00	11
64	00	01	00
65	00	10	00
66	00	11	00
67	01	00	00
68	10	00	00
69	11	00	00

ID NO	BITS 45	67	89
71	00	00	00

3.7.1 Message format. The MINISID III shall be capable of transmitting a Type I or a Type III message as shown in Figures 2A and 2B, respectively. A Type I message consists of a preamble followed by a frame mark, a message ID, the sensor ID, and parity. A Type III message shall be formed by the Type I message followed by T seconds of analog data. The sensor ID portion of the message shall be capable of being set for any of the ID settings shown in Table I.

3.7.2 Preamble (8 bits). The preamble shall be included with each message type and shall consist of eight "zeros".

3.7.3 Frame mark (1 bit). The frame mark shall be included with each message and shall consist of a single bit which shall be a binary one.

3.7.4 Message ID. The two bits preceeding the sensor ID identify the type of message being transmitted. The MINISID III shall be capable of transmitting a Type I or a Type III message as shown in Figures 2A and 2B. A Type I

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message ID is obtained when the lines to bits 2 and 3 in Figure 2A are both at logic "0". A Type III message is obtained when these lines are both at logic "1".

3.7.5 Sensor ID and parity. A parity bit shall be included with the sensor ID code. This parity bit shall follow the bit 9 of the sensor ID and shall be associated with only this ID. The parity bit shall be a "one" if the sum of the bits in the ID is even and shall be a "zero" if the sum is odd.

3.7.6 Bit rate. The bit rate shall be controlled by supplying either a ground or plus 6.3 VDC for selection of either 300 or 75 bits per second respectively, on the bit rate select line.

3.8 Transmitter. The current drain requirement shall be not more than 430 ma at 28 VDC. The transmitter shall deliver a nominal 4 watts, with a minimum of 2 watts for a battery voltage of 22 to 30 VDC, and at least 1 watt for a battery voltage of 20 VDC, when terminated in a 50 ohm resistance load or the very high frequency (VHF) transmitting antenna. The transmitter shall be stabilized on the space frequency and delivering at least minimum power 5 ms or less after receipt of an enable pulse. The transmitter power shall decay at least 10dB within 5 msec, at the end of the transmitter enable gate. The harmonic emission from the transmitter shall be a minimum of 40 decibel (dB) down from the fundamental frequency; spurious emissions shall be a minimum of 40 dB down from the fundamental frequency outside of the emission bandwidth. The transmitter shall be capable of being modulated by signals from below 10 Hz to above 3.6 kHz. All active elements shall be solid-state devices. Carrier frequency error shall not exceed ± 30 parts per million (PPM) under any combination of environmental conditions and within the duty factors specified herein. The transmitter carrier stability shall be ± 5.2 kHz or better. Each Type III message shall contain T seconds of analog data following the ID parity bit. A maximum deviation of 3 kHz ± 10 percent can be expected under normal operating conditions, however, deviations up to 3.6 kHz may be experienced under extreme input levels. A maximum of 5 percent distortion of analog signals will occur where the deviations are limited to ± 3 kHz. The analog frequency response shall be flat within ± 3dB referenced to 100 Hz over the 40-2000 Hz range. The response at 8 kHz shall be down at least 12dB from the 100 Hz reference. The response shall asymptotically roll off at a 12dB/octave rate beyond 8 kHz.

3.8.1 Channel characteristics. The channels will be numbered 1 thru 1920, beginning at a 162.00625 Megahertz (MHz) center frequency, and spaced at 6.25 kHz intervals thereafter. In order to accommodate an inter-channel guard band of 18.75 kHz, only every third channel will be used, beginning with Channel 1 and ending with Channel 1918. This will result in 640 possible VHF channel

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center frequencies for the MINISID III. These channel numbers will be (1+3N) where N varies from 0 to 639. The channel number and frequency equivalents shall be as follows:

Channel	Center Frequency in MHz
1	162.00625
2	162.01250
3	162.01875
4	162.02500
-	-
-	-
-	-
1918	173.98750
1919	173.99375
1920	174.00000

3.8.2 Disable circuit. The transmitter shall contain a nonexplosive, non-pyrotechnic self-disable device that is capable of permanently disabling the transmitter module. The energy required to activate the device is 30,000 ergs at 300 milliampere (ma). The disable device shall not be activated by transients during normal operation but shall be activated by any one or any combination specified in 3.9.5.2.

3.9 SCUE. The SCUE shall consist of the following major components:

(a) Case.
(b) Geophone and cable assembly.
(c) Seismic module assembly.
(d) Code plug assembly.
(e) Antenna.
(f) Disable control assembly.

3.9.1 Geophone and cable module. The geophone and cable assembly shall be as specified in applicable drawings and as described herein. The cable shall be securely attached to the geophone and sensor assembly case and shall be capable of withstanding a 20-pound pull in any direction on the cable without damage.

3.9.2 Seismic module. The seismic module shall be field replaceable without special tools or test equipment and shall weigh less than 14 ounces. The operating power shall be obtained from the 28 VDC battery and from the regulated 6.3 VDC of the encoder. In the event that seismic signals are continuously detected, outputs from the processing circuitry shall occur at 10 seconds (sec) $\pm$ 2 sec intervals.

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3.9.2.1 Seismic bandpass amplifier/processor. The seismic bandpass amplifier shall have a bandpass that is flat within 3dB from 7 to 40 Hz. The response shall be 3dB down from its peak at 45 Hz ± 5 Hz and 9 Hz ± 2 Hz. The high end roll off shall be greater than 9dB/octave for frequencies greater than 200 Hz. The seismic bandpass amplifier fixed gain settings of "HI", "MED" and "LO" sensitivity shall be such that a sinusoidal 14 Hz seismometer input of 30.8, 77.7 and 190.0 millionths of an inch per second peak-to-peak (micro in/sec p-p) respectively shall cause the threshold of the signal processor to be exceeded and result in a MINISID III activation sequence (see 3.6). The gain shall be as specified in the applicable drawings. The "HI" and "MED" gains shall be selected by applying 5.5 to 6.3 VDC to the proper pin in the seismic module connector. "LO" gain shall be selected when no voltage is applied to these pins. The amplified seismic signals within the bandpass of the seismic bandpass amplifier shall be processed and the presence of desired seismic signals shall produce a standard 15 ± 5 ms output pulse having a wave form as shown in Figure 4. The rise and fall time of this output pulse shall be less than 0.3 msec between the 10 percent and 90 percent points. The amplitude shall be 5.5 to 6.3 V as measured into a load of 1,000 ohms.

3.9.3 Code plug. The sensor ID and parity, the seismic module sensitivity, message bit rate, and complementary ID shall be controlled by setting of the code plug. The code plug settings are made by opening one or more possible conduction paths for each bit in the message. A "one" or positive level is +5.5 V to +6.5 V. A "zero" or ground level is 0 to +1 V.

3.9.3.1 Code plug settings. The following settings shall be made by positioning the jumpers located on the code plug:

(a) The sensor ID and parity portions of the message format (see 3.7.1).
(b) The message bit rate (see 3.7.6).
(c) The seismic module sensitivity (see 3.9.2.1).
(d) Complementary ID (CID). The CID shall select the message format in response to a call-up signal from an ancillary device. When set in the "0" position, the ID set on the code plug shall be transmitted. When set in the "1" position, the complement of the ID set on the code plug shall be transmitted. The seismic module ID shall always be the one set on the code plug, regardless of the setting of the CID control.

3.9.4 Antenna. The MINISID III antenna shall present a nominal 50 ohm resistive load and a maximum Voltage Standing Wave Ratio (VSWR) of 2:1 to the transmitter.

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3.9.5 Disable control assembly.

3.9.5.1 External controls. The MINISID III shall contain the following external manual controls. Each switch shall have a mechanical stop to prevent damage to the switch by attempts to turn the switch past the normal operating range.

3.9.5.1.1 Power switch. A three position switch shall be used for the following functions and shall contain a mechanical stop that must be pressed in order to move the switch from the "TEST" to "ARM" position or the "ARM" to "TEST" position.

(a) "OFF" position. Power shall not be applied to any circuitry.
(b) "TEST" position. This switch position shall apply power to all modules except the disable module. A test can be performed on the complete MINISID III without danger of activating the disable function.
(c) "ARM" position. This switch position shall apply power to all module assemblies including the disable module. With the switch in the "ARM" position, the disable circuit shall function in accordance with 3.9.5.2.

3.9.5.1.2 Security switches. The security switches shall be interconnected to allow deactivation of the disable circuit to permit intact recovery of the MINISID III by authorized personnel. The security switches shall permit the power switch to be turned from "ARMED" to "OFF" without activating the disable circuit. The "SAFE" code positions for security switches shall be stamped on the inside of the battery compartment and on the identification tag.

3.9.5.2 Disable system. The disable system shall be capable of permanently disabling the transmitter module. The disable signal pulse shall be as shown in Figure 5. The disable system shall not be activated by transients during normal operation but shall be activated by any one or any combination of the following:

(a) Removal of any of the MINISID III covers.
(b) A battery voltage sensor that shall activate the disable circuit when the battery voltage is reduced to 20.7 VDC + 5 percent.
(c) By manually turning the power switch from "ARMED" to "TEST" with either of the security switches in an "UNSAFE" position.

3.10 MINISID III current drain. Maximum current requirements shall be as follows.

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3.10.1 Current drain for MINISID III subassemblies.

Voltage/Load	Operating mode	
	Test	Arm
28 VDC/Encoder.	0.52 ma	0.52 ma
28 VDC/Disable control assembly.	1.0 microampere(μa)	0.60 ma
28 VDC/Seismic module assembly.	0.48 ma	0.48 ma
6.3 VDC/Seismic module.	60 μa	60 μa
6.3 VDC/Disable control assembly.	10 μa	10 μa

3.10.2 Current drain for complete MINISID III.

Voltage/Load	Operating mode
	Test
28 VDC/Standby	1.0 ma

3.11 Government-furnished property. When the complete MINISID III is being furnished (see 3.1), the Government will furnish the following property in the quantities indicated (see 6.4):

Item No.	Description	Identification	Quantity for each MINISID III
1	Encoder	KY-678/GSQ	1
2	Transmitter	T-1143/GSQ	1
3	Battery	BA-1546/U	1

3.12 Government-loaned property. The Government will lend the following property in the quantities indicated (see 6.5):

Item No.	Description	Identification	Quantity
1	Radio Frequency Monitor Set.	AN/USQ-46 or equivalent.	As specified (see 6.2).
2	Common Module Stack Test Set.	PT-1585	One per contract.

3.12.1 Government-loaned property when only the SCUE is being furnished. When only the SCUE is being furnished (see 3.1), the Government will loan the following property in the quantities specified in 3.1 (see 6.5):

Item No.	Description	Identification
1	Encoder	KY-678/GSQ
2	Transmitter	T-1143/GSQ
3	Battery	BA-1546/U

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3.13 Electromagnetic interference. The electromagnetic interference emission and susceptability characteristics of the MINISID III shall conform to MIL-STD-461; Tests RS01, RS03, CE06, and RE02.

3.14 Burn-in. Each SCUE to be delivered to the Government shall be operated without failure for 48 consecutive hours with 100 activations per day.

3.15 Safety. The equipment shall meet the applicable safety requirements of MIL-STD-454.

3.16 Reliability. The MINISID III shall have a specified Mean-Time-Between-Failure (MTBF) of 10,382 hours assuming 100 activations per day. When specified (see 6.2), reliability testing shall be conducted in accordance with 4.6.3.14.

3.17 Workmanship. Workmanship shall be not less than that specified in MIL-W-27076. All parts, components, and assemblies of the MINISID III including castings, forgings, molded parts, stamping, seals, machined surfaces, and welded parts shall be clean and free from sand, dirt, fins, pits, sprues, scale, flux, weld spatter and other harmful extraneous materials. External surfaces shall be free from burrs and sharp edges.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. The supplier is responsible for insuring that components and materials used are manufactured, examined and tested in accordance with referenced specifications and standards.

4.1.2 Disassembly inspection. Failure of any test or examination of the preproduction model shall be cause for disassembly, in the presence of a Government representative, of the preproduction model to the extent necessary to determine the cause of the failure. Each disassembled part shall be examined in detail for compliance with this specification and referenced drawings in regard to materials, dimensions, tolerances, and workmanship. Parts not complying with such requirements shall be rejected and shall be cause for rejecting the preproduction model. Government-furnished or Government-loaned property shall be excluded from the requirements of this paragraph.

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4.2 Classification of inspections. Inspections shall be classified as follows:

(a) Preproduction inspection (see 4.3).
(b) Initial production inspection (see 4.4).
(c) Quality conformance inspection (see 4.5).
(d) Inspection comparison (see 4.7).
(e) Inspection of preparation for delivery (see 4.8).

4.3 Preproduction inspection.

4.3.1 Examination. The preproduction MINISID III shall be examined as specified in 4.6.1. Presence of one or more defects shall be cause for rejection or cause for performing the inspection specified in 4.1.2.

4.3.2 Tests. The preproduction MINISID III shall be subjected to the tests marked "X" in Column 1 of Table II. Failure of any test shall be cause for performing the inspection specified in 4.1.2.

4.4 Initial production inspection. When specified (see 3.3), one or more initial production MINISID III will be selected at random by the Government from the first MINISID III being produced from production tooling and will be examined for the defects in 4.6.1 and subjected to the tests marked "X" in Column 2 of Table II to determine conformance to the requirements of this specification. The inspection will be performed by the Government at a site selected by the Government. Acceptance of an initial production MINISID III shall not exclude the remaining MINISID III from the quality conformance inspection and acceptance provisions specified in Section 4. In addition to any test specified as part of the initial production test, the Government reserves the right to conduct any and all other tests contained in this specification as part of the initial production test and failure of such additional tests shall have the same effect as failure of those tests specified as initial production tests.

4.4.1 Inspection failure. Failure of an initial production MINISID III to meet any requirement specified herein during and as a result of the inspection specified in 4.4 shall be cause for rejection of the initial production MINISID III and shall be cause for refusal by the Government to continue acceptance of production MINISID III until evidence has been provided by the supplier that corrective action has been taken to eliminate the deficiencies. Correction of such deficiencies shall be accomplished by the supplier at no cost to the Government on MINISID III previously accepted and produced under the contract. Any deficiencies found as a result of the initial production inspection will be considered prima facie evidence that all MINISID III accepted prior to the completion of initial production inspection are similarly deficient unless evidence to the contrary is furnished by the supplier and such evidence is acceptable to the contracting officer.

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4.5 Quality conformance inspection.

4.5.1 Sampling. Sampling for tests shall be in accordance with MIL-STD-105, Inspection Level II.

4.5.2 Examination.

4.5.2.1 Individual. Each MINISID III shall be examined for the defects in 4.6.1. Presence of one or more defects shall be cause for rejection.

4.5.3 Tests.

4.5.3.1 Individual. Each MINISID III shall be subjected to the tests marked "X" in Column 3 of Table II. Failure of any test shall be cause for rejection.

4.5.3.2 Samples. Samples selected in accordance with 4.5.1 shall be subjected to the tests marked "X" in Column 4 of Table II. AQL shall be 1.5 percent defective. No adjustments or substitutions of components shall be made on the MINISID III after completion of the individual tests.

4.6 Inspection procedure.

4.6.1 Examination. The MINISID III shall be examined for the following defects:

101. Protective treatment not as specified.
102. Material not as specified.
103. Printed wiring assemblies and boards not as specified.
104. Fungus and moisture resistance not as specified.
105. Component parts not as specified.
106. Geophone and cable module not as specified.
107. Seismic module not as specified.
108. Antenna not as specified.
109. Carrying strap not as specified.
110. Treatment and painting not as specified.
111. Protective cover not as specified.
112. Identification and instruction marking not as specified.
113. Workmanship not as specified.

4.6.2 Test schedule. The test schedule shall be as shown in Table II.

4.6.3 Tests. Any test failures attributable to Government-furnished equipment shall not constitute failure of any test specified herein.

4.6.3.1 Conditions. Unless otherwise specified herein all tests shall be conducted at an ambient temperature of 75° F ± 5° F with the power on.

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4.6.3.1.1 Test equipment calibration. Calibration of test equipment shall be in accordance with MIL-C-45662.

4.6.3.1.2 Tests, test equipment and test procedures. The list of proposed tests, test equipment and test procedures shall be furnished to the Government within 30 days prior to setting up test equipment. Tests required to determine compliance with the requirements of this specification shall be conducted in accordance with Government approved supplier's test procedures and as specified herein. The same test procedures shall be used for testing a MINISID III or SCUE.

Table II. Test Schedule

Preproduction	Initial Preproduction and Inspection Comparison	Quality Conformance Individual	Quality Conformance Samples	Test	Requirement Paragraph	Test Paragraph
1	2	3	4	5	6	7
X	X	X	X	Activation sequence.	3.6	4.6.3.2
X	X	X	-	Seismic module assembly.	3.9.2	4.6.3.3
X	X	X	X	Encoder.	3.7	4.6.3.4
X	X	X	X	Code plug.	3.9.3	4.6.3.5
X	X	X	X	Transmitter.	3.8	4.6.3.6
X	X	-	-	Antenna.	3.9.4	4.6.3.7
X	X	X	-	Power measurement.	3.8	4.6.3.6.1
X	X	X	-	Disable system.	3.9.5.2	4.6.3.9.1
X	X	-	X	Disable system.	3.9.5.2	4.6.3.9.2, 4.6.3.9.3
X	X	X	X	External controls.	3.9.5.1	4.6.3.10
X	X	-	X	Ancillary devices.	3.6.1	4.6.3.2.1 4.6.3.2.2
X	X	-	-	Vibration.	3.5	4.6.3.11.1
X	X	-	-	Shock.	3.5	4.6.3.11.2
X	X	-	X	High/Low operating temperature.	3.5	4.6.3.12.1
X	X	-	-	High temperature.	3.5	4.6.3.12.2
X	X	-	-	Low temperature	3.5	4.6.3.12.3

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Table II. Test Schedule(Cont'd)

Preproduction	Initial Production and Inspection Comparison.	Quality Conformance Individual	Quality Conformance Samples	Test	Requirement Paragraph	Test Paragraph
1	2	3	4	5	6	7
X	X	-	-	Temperature Shock	3.5	4.6.3.12.1
X	X	-	-	Altitude.	3.5	4.6.3.12.4
X	X	-	X	Humidity.	3.5	4.6.3.12.5
X	X	-	-	Salt fog.	3.5	4.6.3.12.6
X	X	X	X	Immersion.	3.5	4.6.3.12.7
X	-	-	-	Fungus growth (Test A)	3.5	4.6.3.12.8.1
X	X	-	-	Fungus growth (Test B)	3.5	4.6.3.12.8.2
X	X	X	-	Sensor current drain MINISID III sub-assemblies.	3.10.1	4.6.3.8 (a) (b)
X	X	X	-	Sensor current drain MINISID III complete assembly.	3.10.2	4.6.3.8 (c)
X	-	-	-	Electromagnetic interference.	3.13	4.6.3.13
X 3/	X	-	-	Reliability	3.16	4.6.3.14

NOTES: (1) Electrical tests shall be performed after each of the environmental tests or test series.
(2) The immersion test shall be performed after all of the environmental tests except fungus have been completed.
(3) When specified.

4.6.3.2 Activation sequence, MINISID III "SEISMIC ONLY" mode. The MINISID III shall be tested for compliance with 3.6. Compliance shall be verified by use of an AN/USQ-46 or equivalent receiver. The MINISID III seismic processor shall be activated by seismically disturbing the geophone module. Receipt of a valid MINISID III transmission on the correct channel with the proper ID, which is a result of a proper activation sequence (see 3.6) shall constitute proof of proper transmitter performance. Failure to transmit a valid signal (see 3.6.1 and 3.7.1) on the correct channel shall constitute failure of this test.

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4.6.3.2.1 Activation sequence, ancillary device, MAGID T-4 mode. The MINISID III, in conjunction with a signal simulator providing signals equivalent to a MAGID T-4 detector conforming to MIL-D-52776, shall be tested for compliance with 3.6.1. Compliance shall be verified by the use of an AN/USQ-46 or equivalent receiver and the signal simulator. In addition to the signal simulator, an external jumper and test point (TP) shall be provided. The signals and test conditions for testing the MINISID III in conjunction with the signal simulator shall be as shown in Figure 6. The MINISID III geophone shall be seismically disturbed which shall produce a pulse at pin "D" (TP) with characteristics as shown in Figure 4. A positive pulse with the characteristics shown in Figure 6 shall be produced by the signal simulator and applied to the output connector pin "A". This pulse shall activate the encoder and produce a transmission of the message format in accordance with 3.7.1 and 3.9.3. The receipt of a valid MINISID III transmission (see 3.6.1 and 3.7.1) resulting in activations from the signal simulator when placed in the MAGID T-4 mode shall constitute proper performance of the MINISID III ancillary device input requirements. Failure to respond to MINISID III call-up signals and switching control from the ancillary device simulator to produce the valid MINISID III transmission (see 3.6.1 and 3.7.1) shall constitute failure of this test.

4.6.3.2.2 Activation sequence, ancillary device AAU mode. The MINISID III, in conjunction with a signal simulator providing signals equivalent to an AAU detector conforming to MIL-D-52778, shall be tested for compliance with 3.6.1. Compliance shall be verified by use of the stack tester, an AN/USQ-46 or equivalent receiver, and the signal simulator. In addition to the signal simulator, an external jumper and test point shall be provided. The signals and test conditions for testing the MINISID III in conjunction with the signal simulator shall be as shown in Figure 7. The MINISID III geophone shall be seismically disturbed in a manner that produces a series of three pulses in 29 sec $\pm$ 1.5 sec at pin "D" of the output connector. Individual pulses shall be positive pulses with characteristics as shown in Figure 4. This output from the MINISID III shall activate the AAU simulator and shall be observed at test point (TP). The MINISID III transmitter shall respond to the call-up signal generated by the AAU simulator and be modulated for the time period and with an audio analog signal as shown in Figure 7. The receipt of a valid MINISID III transmission followed by the 15 second audio analog signal (see 3.6.1 and 3.7.1) resulting in activations from the signal simulator when placed in the AAU mode shall constitute proper performance of the MINISID III ancillary device input requirement. Failure to respond to MINISID III call-up signals and switching control from the ancillary device simulator to produce the valid MINISID III transmission (see 3.6.1 and 3.7.1) shall constitute failure of this test.

4.6.3.3 Seismic module assembly. The MINISID III shall be tested as follows and nonconformance to 3.9.2 shall constitute failure of this test.

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4.6.3.3.1 Test procedures. This test shall be performed on the MINISID III with the geophone seismically isolated so as to prevent activation of the seismic module.

(a) Supply a 20 Hz continuous wave (CW) signal at the seismic module test input (pin "J", Figure 1) at a level of 3dB ± 0.2dB above the threshold for the particular module installed in the MINISID III under test. A seismic module output pulse (pin D of Figure 1) shall be obtained within 5 seconds or less; and shall continue at 8 to 12 second intervals for a total of four pulses.
(b) Supply a 20 Hz signal in bursts of 0.5 seconds duration every 10 seconds at the seismic module test input at a level 6dB ± 0.2dB above the threshold for the particular module installed in the MINISID III under test. There shall not be a seismic module output pulse for a period of at least four signal bursts.
(c) Check output pulse.
(d) Tap the geophone gently to cause at least three seismic module output pulses. Failure to generate three pulses at 10 ± 2 sec intervals shall constitute failure of this test.

4.6.3.4 Encoder. The encoder shall be tested using the Government-loaned stack test set. Failure to generate a properly encoded message in accordance with 3.7 shall constitute failure of this test.

4.6.3.5 Code plug. Set the ID code at 29 and 71; at each code, operate the MINISID III at the various gains, bit rates, and parities; ancillary equipment shall be used to determine the CID. Assure that gain setting changes are actually accomplished. Nonconformance to 3.9.3 shall constitute failure of this test.

4.6.3.6 Transmitter. The transmitter shall be tested using the Government-loaned stack test set (see 3.12). Nonconformance to 3.8 shall constitute failure of this test.

4.6.3.6.1 Power measurement. Activate the complete MINISID III by tapping on the geophone. Measure the RF power into the antenna with a battery voltage of 22 to 28 VDC. Failure of the MINISID III to deliver not less than 2 watts RF power to the antenna shall constitute failure of this test.

4.6.3.7 Antenna VSWR. The VSWR of the antenna shall be measured at 168 Megahertz MHz), 162 MHz, and 174 MHz. A VSWR greater than 2:1 (see 3.9.4) shall constitute failure of this test.

4.6.3.8 Sensor current drain. The current drain of the MINISID III and subassemblies shall be measured. Current drain of subassemblies may be measured prior to installation into the case.

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(a) Supply 28 VDC ± 0.5 VDC and 6.3 VDC, ±0.2 VDC simultaneously to the seismic module assembly (pins 21 and 20 of the module, respectively). Measure the current drain.
(b) Supply 28 VDC ± 0.5 VDC and 6.3 VDC ± 0.2 VDC to the disable control assembly (pins 8 and 25 of the module, respectively). Measure the current drain.
(c) Supply 28 VDC ± 0.5 VDC to the MINISID III through the external connector. Measure the current drain under standby conditions (not transmitting) for both the "TEST" and "ARM" mode.

Nonconformance to 3.10 shall constitute failure of these tests.

4.6.3.9 Disable system.

4.6.3.9.1 Disable system test at 75° F ± 5° F. The disable circuit shall be tested utilizing a dummy load in place of the transmitter to prove compliance with the requirements of 3.9.5.2. The dummy load shall contain a 6 ohm resistor between pins 47 and 52 to simulate the squib in the transmitter. The disable pulse at pin 52 shall be monitored with the oscilloscope. Nonconformance to 3.9.5.2 shall constitute failure of this test.

4.6.3.9.2 Disable system test at 32° F. Repeat the test of 4.6.3.9.1 in an ambient temperature of 32° F ± 5° F. Nonconformance to 3.9.5.2 shall conconstitute failure of this test.

4.6.3.9.3 Disable system test at 145° F. Repeat the test of 4.6.3.9.1 in an ambient temperature of 145° F ± 5° F. Nonconformance to 3.9.5.2 shall constitute failure of this test.

4.6.3.10 External controls. Compliance with the performance requirements of 3.9.5.1 shall be verified by the following tests.

4.6.3.10.1 Power switch. The power switch shall be tested to prove compliance with the requirements of 3.9.5.1.1. Failure of the switch to apply power, remove power, or arm the unit (see 3.9.5.1.1) shall constitute failure of this test.

4.6.3.10.2 Security switch. The two security switches shall be checked in all possible combinations of position to confirm that there is only one safe position (see 3.9.5.1.2) and that this is the position marked on the identification tag. Failure to confirm that one and only one safe position exists and agrees with the marking inside the battery compartment and on the identification tag (see 3.9.5.1.2) shall constitute failure of this test.

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4.6.3.11 Physical tests. Unless otherwise specified herein the transmitter, encoder, and battery shall be installed in the MINISID III during these tests. After all physical tests, the MINISID III shall be subjected to all tests specified in Column 3 of Table II, except immersion (4.6.3.12.7) and sensor current drain (4.6.3.8 (a) and (b)); and inability to pass any of these tests or evidence of damage (see 6.7.1) shall constitute failure of the shock test.

4.6.3.11.1 Vibration (power off). The MINISID III shall be subjected to sinusoidal vibration in its three mutually perpendicular axes consisting of logarithmic cycling from 5 to 200 to 5 Hz. The sweep rate shall be 5 to 200 to 5 Hz in 12 minutes. Vibratory inputs to the test item shall be 1 inch double amplitude constant displacement from 5 to 7 Hz, 2.5 G's peak constant acceleration from 7 to 92 Hz, 0.0058 inches double amplitude constant displacement from 92 to 110 Hz, and 3.5 G's peak constant acceleration from 110 to 200 Hz. Total test times shall be 84 minutes per axis of vibration. Operational checks shall be performed after each axis of vibration. The MINISID III shall be tested as specified in 4.6.3.2. Failure to generate a valid transmission (see 3.6.1 and 3.7.1) or any evidence of damage (see 6.7.1) shall constitute failure of this test.

4.6.3.11.2 Shock. The MINISID III shall be subjected to the following shock tests in accordance with MIL-STD-810. After each of the following shock tests, the MINISID III shall be tested as specified in 4.6.3.2:

(a) (Power off). With the MINISID III out of its shipping container, test in accordance with Method 516.1, Procedure I, Figure 516.1-2 Amplitude b, time duration, 11 seconds. Failure of the MINISID III to generate a valid transmission (see 3.6.1 and 3.7.1) or any evidence of damage (see 6.7.1) shall constitute failure of this test.

(b) (Power off). With the MINISID III in its shipping container, test in accordance with Method 516.1, Procedure II. Failure of the MINISID III to generate a valid transmission (see 3.6.1 and 3.7.1) or any evidence of damage (see 6.7.1) shall constitute failure of this test.

(c) (Power on). With the MINISID III out of its shipping container, test in accordance with Method 516.1, Procedure V. Failure of the MINISID III to generate a valid transmission (see 3.6.1 and 3.7.1) or any evidence of damage (see 6.7.1) shall constitute failure of this test.

4.6.3.12 Environmental tests. Unless otherwise specified herein, the transmitter, encoder and battery shall be installed in the MINISID III. After completion of tests in accordance with 4.6.3.12.1 thru 4.6.3.12.7; the MINISID III shall be subjected to all tests specified in Column 3 of Table II, except immersion (4.6.3.12.7) and sensor current drain (4.6.3.8 (a) and (b); and inability to pass any of these tests or evidence of damage (see 6.7.1) shall constistute failure of the environmental tests.

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4.6.3.12.1 High/low operating temperature and temperature shock. The MINISID III at operating temperature ranges, shall be subjected to the following high and low temperature cycle shock tests. Failure of the MINISID III to generate a valid transmission (see 3.6.1 and 3.7.1) or any evidence of damage (see 6.7.1) shall constitute failure of this test.

(a) Place MINISID III in test chamber and activate. Raise the temperature to +145° F and maintain for 4 hours. Perform an activation sequence test (see 4.6.3.2) and then remove the battery, transmitter, and encoder.
(b) Raise the temperature to +160° F and maintain for 4 hours.
(c) Reduce the temperature at a rate of not less than 45° F per minute to -65° F and maintain for 4 hours.
(d) Raise the temperature at a rate of not less than 45° F per minute to +160 F and maintain for 4 hours.
(e) Reduce the temperature to +32° F and maintain for 4 hours. At this point reinstall the battery, transmitter, and encoder and activate the MINISID III. Perform an activation sequence test (see 4.6.3.2).
(f) Raise the temperature to ambient room temperature and maintain for 1 hour. At the conclusion of this test the MINISID III shall be subjected to an activation sequence test (see 4.6.3.2).

4.6.3.12.2 High temperature. The MINISID III shall be tested in accordance with MIL-STD-810, Method 501, Procedure I. The highest operating temperature shall be 145° F. Perform an activation sequence test (see 4.6.3.2). Failure of the MINISID III to generate a valid transmission (see 3.6.1 and 3.7.1), or any evidence of damage (see 6.7.1) shall constitute failure of this test.

4.6.3.12.3 Low temperature. The MINISID III shall be tested in accordance with MIL-STD-810, Method 502, Procedure I. The lowest temperature shall be -65° F maintained for 8 hours. The operating temperature shall be 32° F for 8 hours. Perform an activation sequence test (see 4.6.3.2). Failure of the MINISID III to generate a valid transmission (see 3.6.1 and 3.7.1), or any evidence of damage (see 6.7.1), shall constitute failure of this test.

4.6.3.12.4 Altitude (power off). The MINISID III shall be tested in accordance with MIL-STD-810, Method 500, Procedure I except that the operating altitude shall be 14,000 ft. Perform an activation sequence test (see 4.6.3.2). Failure of the MINISID III to generate a valid transmission (see 3.6.1 and 3.7.1), or any evidence of damage (see 6.7.1) shall constitute failure of this test.

4.6.3.12.5 Humidity. With the MINISID III out of its shipping container and the transmitter covers removed, test in accordance with MIL-STD-810,

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Method 507, Procedure I with the following exception: Sample MINISID III (see 4.6.2) shall be subjected to only 5 cycles of Steps 2, 3, and 4 of Procedure I. Perform an activation sequence test (see 4.6.3.2). Failure of the MINISID III to generate a valid transmission (see 3.6.1 and 3.7.1), or any evidence of damage (see 6.7.1) shall constitute failure of this test.

4.6.3.12.6 Salt fog (power off). The MINISID III with all covers, seals and protective devices removed as required for maintenance, settings or adjustments, shall be tested in accordance with MIL-STD-810, Method 509, Procedure I. Perform an activation sequence test (see 4.6.3.2). Failure of the MINISID III to generate a valid transmission (see 3.6.1 and 3.7.1), or any evidence of damage (see 6.7.1) shall constitute failure of this test.

4.6.3.12.7 Immersion (power off). The MINISID III SCUE shall be subjected to the leakage test in accordance with MIL-STD-810, Method 512, Procedure I, except that the temperature difference between the water and the SCUE at the time of immersion shall not be less than 10° C. The water shall be cooler. The top of the case constitutes the uppermost point of the unit. Any evidence of leakage as indicated by bubbles emanating from the exterior of any component shall constitute failure of this test. (Bubbles which are the result of entrapped air on the exterior of component parts shall not be considered a leak.) Within 30 minutes and 240 minutes after removal from the water the SCUE shall be visually inspected for entrance of water, examined for damage (see 6.7.1) and tested in accordance with 4.6.3.2. Any evidence of damage (see 6.7.1) or inability to generate a valid transmission (see 3.6.1 and 3.7.1) shall constitute failure of this test.

4.6.3.12.8 Fungus growth (power off).

4.6.3.12.8.1 Fungus growth (Test A). The MINISID III shall be placed in the tropical testing chamber (see 6.6) and subjected to exposure under simultated cyclic tropical conditions prevailing therein for a period of 90 days. Environmental conditions within the chamber shall be as follows:

> The tropical testing chamber shall be operated on a daily temperature humidity cycle typical of a tropical area. For an 8-hour period, the temperature shall be held at 85° F and the relative humidity at 70 percent. During the next 8 hours of the cycle, the temperature shall be gradually lowered to 75° F, and the relative humidity raised to 95 percent. The temperature and relative humidity shall be then kept at 75° F and 95 percent for a 4-hour period, followed by a gradual change to 85° F and 70 percent during the next 4-hour period, which completes the cycle.

The chamber atmosphere shall be heavily charged with spores of various species of fungi and provide continuous inoculation of exposed items and materials to

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such an extent that artificial prior inoculation with fungus spores is unnecessary. The following list is representative of the fungi to be present in the chamber:

Acremonium alternatum	Memmoniella echinata
Ascotricha xylina	Mucor sp.
Aspergillus flavus	Myrothecium verrucaria
Aspergillus niger	Nigrospora sphaerica
Aspergillus oryzae	Penicillium citrinum
Aspergillus terreus	Penicillium luteum
Aspergillus versicolor	Penicillium purpurogenum
Chaetomium bostrychodes	Penicillium stoloniferum
Chaetomium indicum	Pollularia sp.
Chaetomium globosum	Rhizopus nigricans
Coniothyrium sp.	Spicaria sp.
Curvularia sp.	Stachbotrys atra
Fusarium sp.	Trichoderma viride

Upon removal from the chamber after the 90-day exposure period, the MINISID III shall be examined for damage (see 6.7.1). The MINISID III shall then be subjected to all tests specified in Column 3 of Table II, except immersion (4.6.3.12.7) and sensor current drain (4.6.3.8 (a) and (b)). Inability to pass any of these tests or evidence of damage (see 6.7.1) shall constitute failure of the fungus growth test.

4.6.3.12.8.2 Fungus growth (Test B). The MINISID III shall be tested in accordance with MIL-STD-810, Method 508, Procedure I. After conclusion of test, the MINISID III shall be examined for damage (see 6.7.1). The MINISID III shall then be subjected to all tests specified in Column 3 of Table II, except immersion (4.6.3.12.7) and sensor current drain (4.6.3.8 (a) and (b)). Inability to pass any of these tests or evidence of damage (see 6.7.1) shall constitute failure of the fungus growth test.

4.6.3.13 Electromagnetic interference characteristics. One or more MINISID III shall be tested in accordance with MIL-STD-461 Tests RS01, RS03, CE06 and RE02 only. The supplier shall furnish the contracting officer the test plans and test reports in accordance with MIL-STD-461, except that the time interval for submission of reports shall be as specified (see 6.2). Upon approval of the reports by the contracting officer and provided all other requirements of the specification are met, MINISID III meeting all requirements as specified herein shall be used as models for all production runs. Disapproval of any report shall constitute failure of this test.

4.6.3.14 Reliability test. When specified (see 3.16), the MINISID III shall be tested in accordance with MIL-STD-781, Test Plan IVA using the MTBF

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specified in 3.16. During this period of operation the MINISID III shall be subjected to 100 activations per day. When this reliability test is to be conducted, the time frame for conducting this test, the number of MINISID III units to be tested and who shall perform the tests shall be as specified (see 6.2). The test samples shall not be exposed to the shock, salt fog, and fungus growth prior to the reliability testing. Items selected for reliability testing shall have been burned-in as specified in 3.14, and shall have passed the quality conformance inspection specified in 4.5. The performance of the MINISID III battery shall not be tested. Periodic performance checks (see 4.6.3.2) shall be made throughout the reliability test. The tests shall be made under ambient environmental laboratory conditions. Nonconformance to 3.16 shall constitute failure of this test.

4.6.3.14.1 Parameters to be measured. The parameters to be measured and frequency of measurement during testing shall be as follows:

Activation sequence (see 4.6.3.2).	(daily)
Self-disable (see 4.6.3.9).	(weekly)
External controls (see 4.6.3.10).	(weekly)

4.7 Inspection comparison. The Government may select MINISID III at any time during the contract production period and subject them to the examination specified in 4.6.1 and the tests marked "X" in Column 2 of Table II to determine that the quality of the selected MINISID III is equal to the quality standards established during the initial production inspection. The inspection comparison will be performed by the Government at a site selected by the Government and will not include previously inspected preproduction models and initial production MINISID III. In addition to any test specified as part of the inspection comparison, the Government reserves the right to conduct any and all other tests contained in this specification as part of the inspection comparison. Failure of such additional tests shall have the same effect as failure of those tests specified as inspection comparison.

4.7.1 Inspection failure. Failure of any MINISID III to meet any requirement specified herein during and as a result of the examination and tests specified in 4.7 shall be cause for rejection and refusal by the Government to continue acceptance of production MINISID III until evidence has been provided by the supplier that corrective action has been taken to eliminate the deficiencies. Correction of such deficiencies shall be accomplished by the supplier at no cost to the Government on MINISID III produced and accepted under the contract since approval of the initial production MINISID III. Any deficiencies found as a result of inspection comparison shall be considered prima facie evidence that all MINISID III accepted after approval of the initial production MINISID III are similarly deficient unless evidence to the contrary is furnished by the supplier and such evidence is acceptable to the contracting officer.

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4.8 Inspection of preparation for delivery.

4.8.1 Quality conformance inspection of pack.

4.8.1.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.8.1.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.8.1.3 Examination. Sample selected in accordance with 4.8.1.2 shall be examined for the following defects. AQL shall be 2.5 percent defective.

114.	Materials, methods, and containers not as specified for Level A or B. Each incorrect material, method, or container shall be considered one defect.
115.	Components not cushioned within the box as specified for Level A.
116.	Box not closed and sealed by method as specified for Level A.
117.	Quantities packed together exceed the size or weight limitation specified for Level A, B or C.
118.	Marking missing, illegible, incorrect, or incomplete.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be Level A or C, as specified (see 6.2).

5.1.1 Level A. Each complete MINISID III shall be preserved and packaged in accordance with MIL-P-116, Method IIb. The inner container shall be the suppliers molded expanded polystyrene container or a box conforming to PPP-B-636, Type CF, Class Domestic, Variety SW, Grade 125. Two MINISID III units may be packaged together when the suppliers container is used and the container designed in this manner. The contents shall be cushioned and secured within the box or container in a manner to prevent movement or damage. The outer container shall conform to PPP-B-636, W6c, style optional. Box closure and sealing of the inner container shall be by the most appropriate method. The exterior box shall be closed and sealed in accordance with the appendix to the box specification as specified for Method V.

5.1.2 Level C. Each complete MINISID III shall be packaged, cushioned, and secured in a molded expanded polystyrene container or fiberboard box that will afford protection against mechanical or physical damage from the supplier to the initial destination. Two MINISID III units may be packaged together when the container used is of molded expanded polyethylene and designed in this manner.

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5.2 Packing. Packing shall be Level A, B, or C, as specified (see 6.2).

5.2.1 Level A. The MINISID III units preserved and packaged as specified in 5.1 shall be packed together in a close-fitting box conforming to PPP-B-601, Overseas Type, style optional, Grade B, in quantities not to exceed 200 pounds. Closure and strapping shall be in accordance with the appendix to the box specification.

5.2.2 Level B. The MINISID III units, preserved and packaged as specified in 5.1, shall be packed as specified in 5.2.1 for Level A except boxes shall be Domestic Type. As an alternate boxes conforming to PPP-B-636, V3c, V11c, V13c, or V15c may be used when the quantity to be packed does not exceed the gross weight or size limitation of the box. Box closure and strapping shall be in accordance with the appendix to the applicable box specification.

5.2.3 Level C. The MINISID III, preserved and packaged as specified in 5.1, shall be packed to assure carrier acceptance and safe delivery to destination at lowest ratings in containers complying with Uniform Freight Classification rules or with National Motor Freight Classification rules.

5.3 Marking. In addition to any special marking required by the contract or purchase order, each unit package and shipping container shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The intended use of the MINISID III is to detect seismic signals, and magnetic and acoustic signals when used with an ancillary device, resulting from movement of vehicles and personnel on foot. The ancillary devices used are as follows:

(a) Detector, Anti-Intrusion: DT-509/GSQ (MAGID T-4) conforming to MIL-D-52776(ME).
(b) Detector, Anti-Intrusion: DT-383/GSQ (AAU) conforming to MIL-D-52778(ME).

The MINISID III is a hand-emplaced, self-contained, anti-intrusion alarm set used for detecting the presence of seismic disturbances by performing a signal signature analysis in order to isolate those disturbances that have specific characteristics. When a disturbance of this unique characteristic is detected, an RF transmitter is activated and emits an identification (ID) code. The code identifies the specific MINISID III and announces the detection of a seismic signal. When used in conjunction with an ancillary magnetic or acoustic detector, the MINISID III transmitter can be activated by an input signal from the ancillary detector used singularly or in conjunction with the seismic signal MINISID III processor.

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6.1.1 Implant parameter. The MINISID III employs an external seismic detector. It is capable of being implanted up to 24 inches from the MINISID III case. The MINISID III performs as described herein while resting on top of the ground or while implanted up to 2 inches above the top of the case. Also, the MINISID III will not be damaged by exposure to sustained seismic activation when covered by salt water to a height of 3 feet above the top of the case. The MINISID III is recoverable, such that it can be removed after hand implant for renewal of battery, re-setting of identification code, changing of transmitter frequency, maintenance, or relocation to another area for implant.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) When complete MINISID III shall be furnished (see 3.1).
(c) The quantity of encoders, transmitters, and batteries that the Government will lend to facilitate testing (this option is applicable when only the SCUE is being furnished) (see 3.1).
(d) Time frame required for submission of preproduction model and number of MINISID III required (see 3.2).
(e) When the Government will conduct any or all of the preproduction model examination and tests. When the Government will conduct some but not all of the preproduction examination and tests, the contracting officer should specify which examination and tests will be conducted by the Government and which examination and tests shall be conducted by the supplier (see 3.2).
(f) When initial production inspection is required, and number of MINISID III to be furnished (see 3.3).
(g) The quantity of receivers that will be loaned by the Government (see 3.12).
(h) When reliability testing is required (see 3.16).
(i) Time frame required for submission of electromagnetic interference test reports (see 4.6.3.13).
(j) When reliability test is required, the time frame for conducting the reliability test, the number of MINISID III units to be tested, and who shall perform the test (see 3.16 and 4.6.3.14).
(k) Level of preservation and packaging and level of packing required (see 5.1 and 5.2).

6.3 Preproduction model. Any changes or deviations of production MINISID III units from the approved preproduction model during production will be subject to the approval of the contracting officer. Approval of the preproduction model will not relieve the supplier of his obligation to furnish MINISID III units conforming to this specification.

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6.4 Government-furnished property. The contracting officer should arrange to furnish the property specified in 3.11, when applicable (see 6.2).

6.5 Government-loaned property. The contracting officer should arrange to loan the property specified in 3.12 and 3.12.1.

6.6 Tropical test chamber. A chamber conforming to the chamber specified in 4.6.3.12.8.1 may be available at USAMERDC, Ft. Belvoir, Virginia. The contracting officer should arrange to make the chamber available for testing the MINISID III, when applicable.

6.7 Definitions.

6.7.1 Damage. Damage is deformation, corrosion, loosening of parts, breakage, or change of fit of any component or part, or any physical condition resulting in noncompliance of the MINISID III with the provisions of this specification.

Custodian:

Army - ME

Preparing activity:

Army - ME

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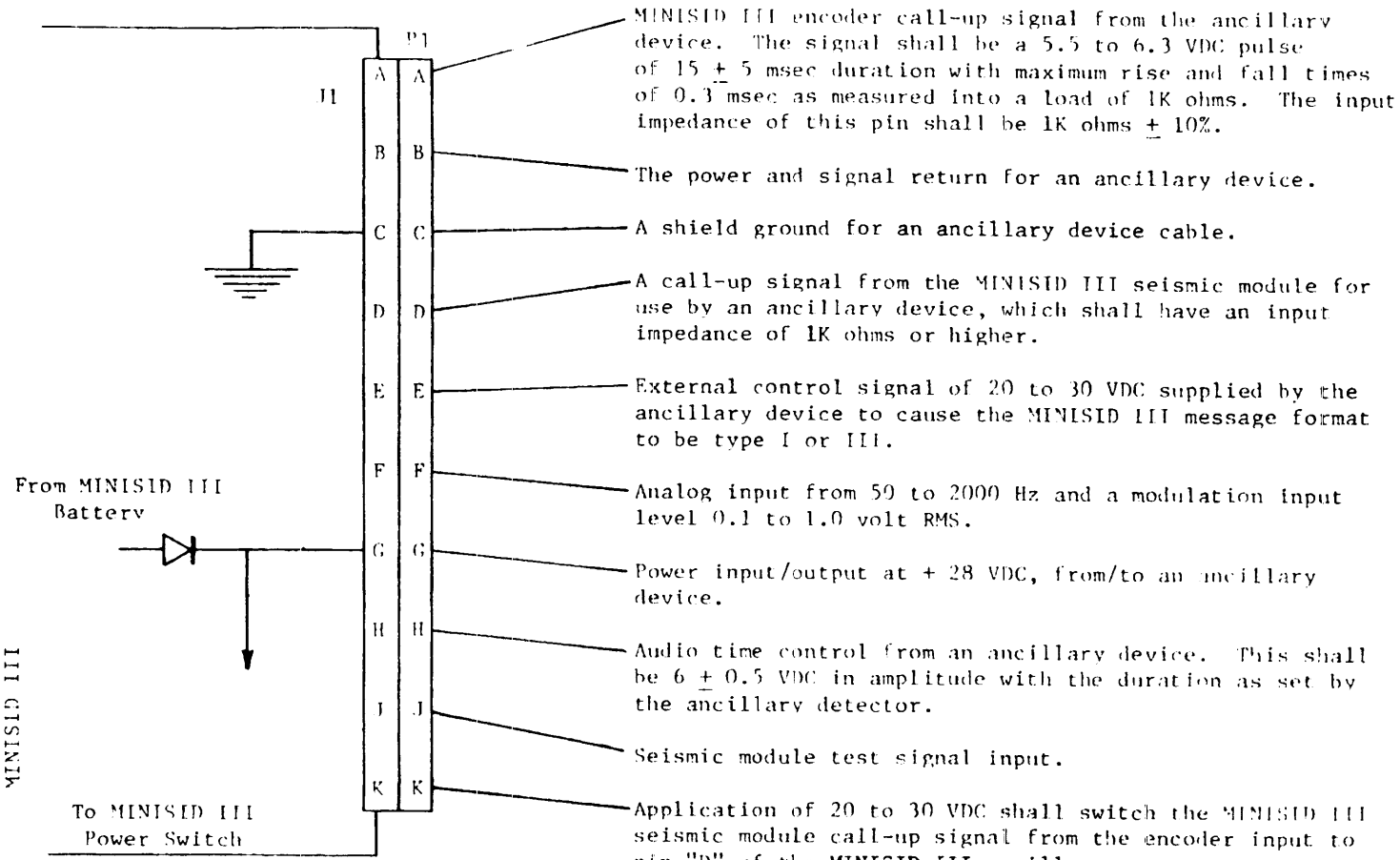


Figure 1. MINISID III/ANCILLARY DEVICE INTERCONNECTION

X-2858

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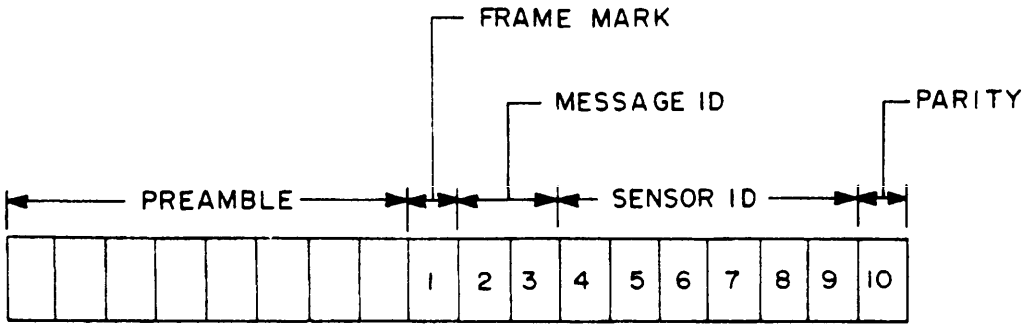


FIGURE 2A. TYPE I MESSAGE FORMAT

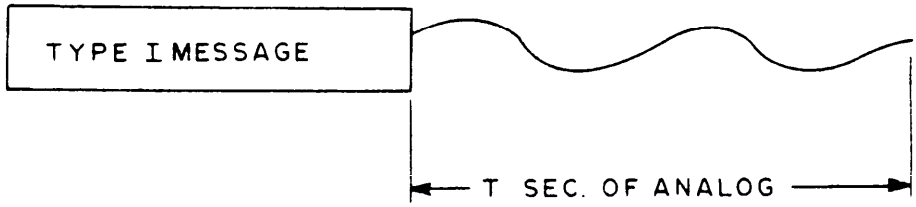
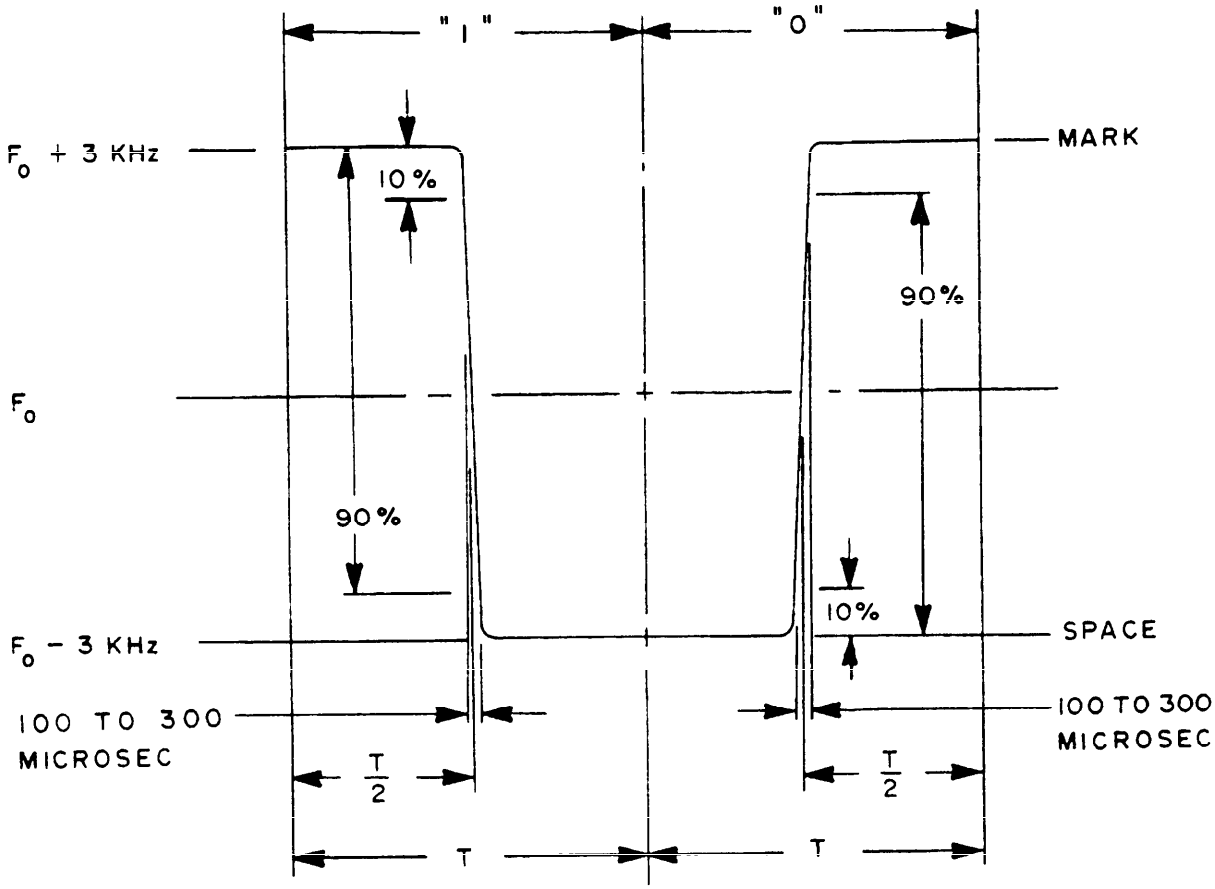


FIGURE 2B. TYPE III MESSAGE FORMAT

X 2859

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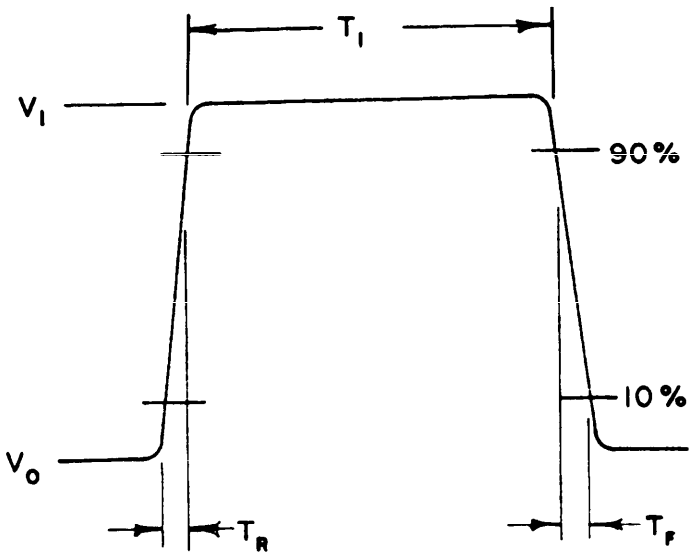
T = 3.33 MS FOR 300 BPS BIT RATE ± 3%

T = 13.33 MS FOR 75 BPS BIT RATE ± 3%

FIGURE 3. MESSAGE BIT STRUCTURE

X-2860

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WHERE V_0 = 0.3 ± 0.3 VOLTS DC

V_1 = +5.5 TO 6.3 VOLTS DC

T_1 = 15 ±5 MILLISECONDS

T_R = LESS THAN 0.3 MILLISECONDS

T_F = LESS THAN 0.3 MILLISECONDS

FIGURE 4. REQUIRED OUTPUT WAVEFORM

X-2862

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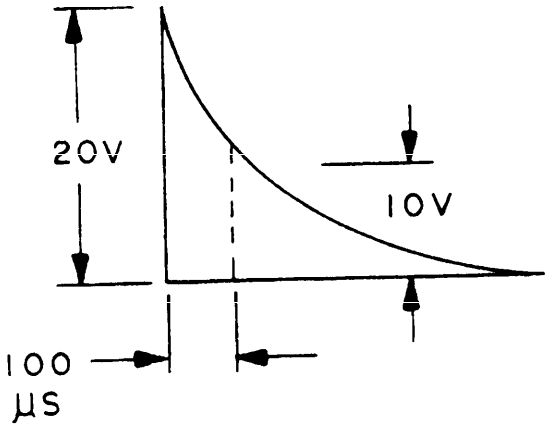


FIGURE 5. DISABLE PULSE WAVEFORM

X 2863

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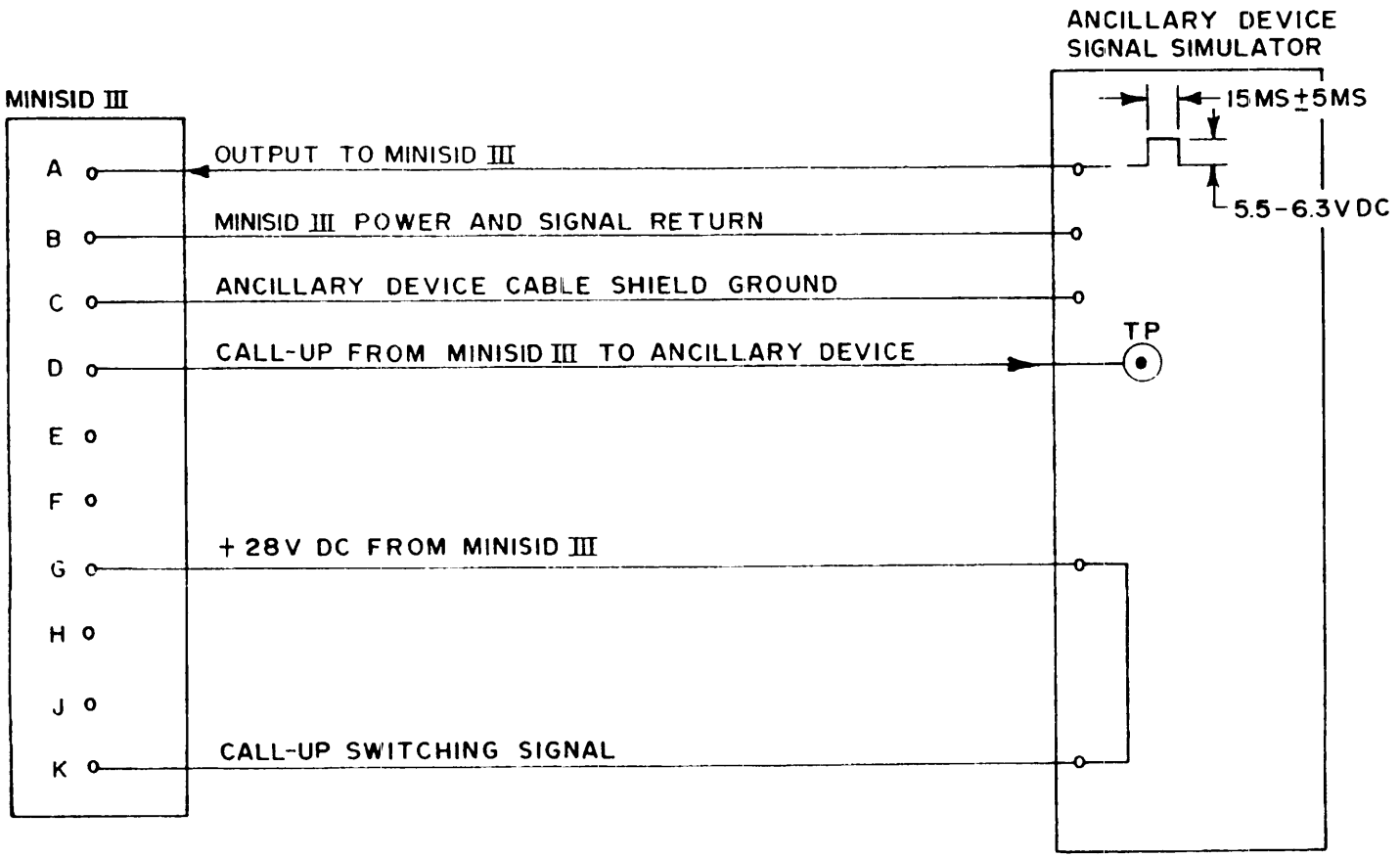


FIGURE 6. PROPOSED MAGID T-4 SIGNAL SIMULATOR TEST FIXTURE

X-2864

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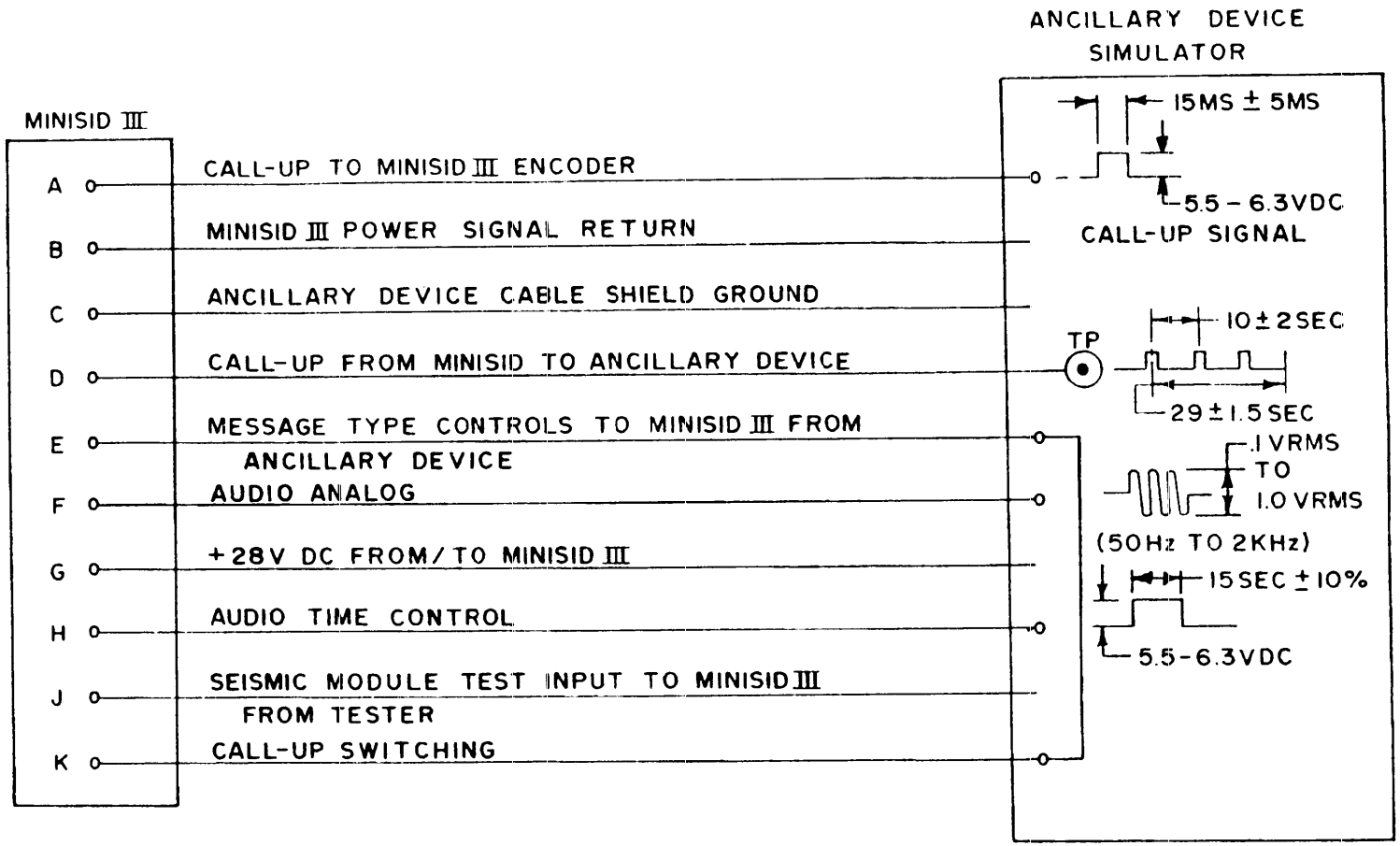


FIGURE 7. PROPOSED AAU SIGNAL SIMULATOR TEST FIXTURE

X-2865

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OMB Approval
No. 22-R255

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DOCUMENT IDENTIFIER AND TITLE MIL-A-52839(ME) Alarm Set, Anti-Intrusion, Restricted Area: MINISID III, AN/GSQ-154

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	MATERIAL PROCURED UNDER A ☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

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   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?
   ☐ YES ☐ NO *(If "Yes", in what way?)*

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